

YICHEN GUO

M.Eng. (by Research), Computer Science and Engineering
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GitHub | Google Scholar | LinkedIn

Seeking PhD opportunities starting in Spring or Fall 2027.
Open to research discussions, recommendations, and potential collaborations.

Research Interests

Vision-language models: efficient inference, visual token reduction, and hallucination mitigation.
World models: action-conditioned generation, action following, and policy learning.
AI agents: post-training and harnesses for finance and empirical research.

Education

Nanyang Technological University Sep 2025 - Jun 2027
M.Eng. (by Research), Computer Science and Engineering; expected graduation in June 2027.

University of Electronic Science and Technology of China Sep 2021 - Jun 2025
B.Eng., Electronic Information Engineering

Research & Professional Experience

GIFTS, Nanyang Technological University Apr 2026 - Present
Research Affiliate, DEFT Lab | Supervisor: Prof. Lin William Cong
Research AI agents for finance and empirical studies. Contribute to InferenceNet evaluation pipelines and reproducibility audits.

HMI Lab, Peking University Jul 2025 - Present
Research Assistant, School of Computer Science | Supervisor: Prof. Shanghang Zhang
Study efficient and reliable vision-language models, including STAR-Pro token pruning and hallucination mitigation; contribute to robotic world-model and action-following research.

Binance Jun 2026 - Present
Data Science Intern, Remote | Supervisor: Harvin Wang
Work on AI trading agents and trading algorithms. Support agent post-training with supervised fine-tuning and reinforcement learning, and harnesses for tool use, execution, and evaluation.

Tencent LightSpeed Studios Aug - Dec 2025
Data Scientist Intern, Singapore | Supervisor: Don Li
Conducted LLM post-training for game-specific tasks and built multi-agent economic simulations.

Harvard AI and Robotics Lab, Harvard University Sep - Nov 2025
Visiting Student, Remote | Supervisor: Prof. Mengyu Wang
Studied reinforcement learning for financial agents and multimodal visual-memory reasoning for medical models, including memory-enhanced post-training.

Publications & Projects

Full publication list and resources: easonai-5589.github.io/publications/

Selected Publications

STAR-Pro: Stage-Wise Token Adaptive Reduction with Progressive Refinement for Efficient Large Vision-Language Models

Yichen Guo, Tinghao Wang, Qizhe Zhang, Lingbei Meng, Yuan Zhang, Jiajun Cao, Hao Jiang, Chenwei Wu, Jixian Wu, Sixiang Chen, Tao Luo, Hongyang Cheng, Kai Tang, Chenxi Li, Renyuan Li, Xiande Huang, Wenya Wang, Shanghang Zhang
Under review, 2026 | Co-first author

Contributed to adaptive-to-progressive visual token pruning design and evaluation.

Do Robotic World Models Really Follow Actions? Diagnosing and Aligning Action-Conditioned Generation for Policy Learning

Sixiang Chen, Jiaming Liu, Jixian Wu, Yichen Guo, Tinghao Wang, Siyuan Qian, Hao Chen, Jiajun Cao, Jian Tang, Shanghang Zhang
Preprint, 2026 | Core contributor

Contributed to WorldEcho / WorldSync research on robotic world models and action following.

SafeDojo: Safe Reinforcement Learning for VLA via Interactive World Model

Kai Tang, Peidong Jia, Zhong Chu, Jixian Wu, Rui Ma, Jiajun Cao, Fangyuan Zhao, Sixiang Chen, Yichen Guo, Xiaowei Chi, Chun-Kai Fan, Kevin Zhang, Jinchang Xu, Fubing Yang, Weishi Mi, Xiaozhu Ju, Jian Tang, Shanghang Zhang
CoRL 2026 - Accepted | Co-author

RegimeVGGT: Layer-Wise Spatially Preserving Redundancy Removal for Visual Geometry Grounded Transformer

Jinhao You, Shuo Lyu, Zhuohang Lyu, Tanxuan Li, Zibo Zhao, Jiaxiang Hu, Kai Tang, Yichen Guo
Under review, 2026 | Co-author

Additional Manuscripts

MAP: Mitigating Hallucinations in Large Vision-Language Models with Map-Level Attention Processing

Manuscript, 2026 | Equal contribution

SeeMe: Mitigating Hallucinations in Large Vision-Language Models through Effective Visual Token Engineering

Manuscript, 2026 | Co-author

Mitigating Hallucinations via Inter-Layer Consistency Aggregation in Large Vision-Language Models

Manuscript, 2025 | Co-author

STAR: Stage-Wise Attention-Guided Token Reduction for Efficient Large Vision-Language Models Inference

Manuscript, 2025 | First author

Earlier STAR manuscript; the current preprint is STAR-Pro.

Research Project

InferenceNet - AI agents for empirical research

A benchmark and research initiative for empirical studies in economics and finance. My work focuses on evaluation pipelines and reproducibility audits. The project page provides task examples, resources, and a research leaderboard.

[Project page](#) | [Dataset](#) | [Paper forthcoming](#)